SURDS

A surd is an irrational number expressed in a root form that can not be simplified into a whole number or a simple fraction.

$\sqrt{a} = a^{1/2}$ $\qquad$ $\sqrt[3]{a} = a^{1/3}$

$\sqrt{5} = (5)^{1/2}$ $\qquad$ $\sqrt[4]{a} = a^{1/4}$

$\sqrt{51} = (51)^{1/2}$ $\qquad$ $\sqrt[5]{a} = a^{1/5}$

- $\sqrt[3]{49} = \sqrt[3]{(7^2)} = (7^2)^{1/3} = 7^{\frac{2}{3}}$
- $\sqrt[4]{343} = (343)^{1/4} = (7^3)^{1/4} = 7^{3\times\frac{1}{4}} = 7^{3/4}$
- $\sqrt{1331} = (1331)^{1/2} = (11^3)^{1/2} = 11^{3\times\frac{1}{2}} = 11^{3/2}$

1.) $a^{\frac{m}{n}} = a^{m\times\frac{1}{n}} = (a^m)^{1/n} = \sqrt[n]{a^m}$

$7^{\frac{2}{3}} = 7^{2\times\frac{1}{3}} = \sqrt[3]{7^2} = \sqrt[3]{49}$

$7^{\frac{3}{2}} = 7^{3\times\frac{1}{2}} = (\sqrt{7})^3 = \sqrt{7^3} = \sqrt{343}$

$\downarrow$

$\sqrt{7\times7\times7} = \sqrt{7}\cdot\sqrt{7}\cdot\sqrt{7}$

$= (\sqrt{7})^3$

2.) $(ab)^n = a^n \cdot b^n$

$(ab)^{1/3} = a^{1/3} \cdot b^{1/3}$

$\sqrt[3]{ab} = \sqrt[3]{a} \cdot \sqrt[3]{b}$

$\sqrt[3]{4\times7} = \sqrt[3]{4} \cdot \sqrt[3]{7}$

$a^n \cdot b^n = ab^n$

$a^{1/3} \cdot b^{1/3} = ab^{1/3} = \sqrt[3]{ab}$

$\sqrt{a} \cdot \sqrt{b} = \sqrt{ab}$

$\sqrt[5]{8} \cdot \sqrt[5]{3} = \sqrt[5]{8\times3}$

$\sqrt{17} \cdot \sqrt{5} = \sqrt{17\times5} = \sqrt{85}$

3) Multiplication | Division

- $\sqrt{a}\cdot\sqrt{b} = \sqrt{ab}$

$\Rightarrow \sqrt{2}\cdot\sqrt{5} = \sqrt{10}$ $\Rightarrow \sqrt{12}\cdot\sqrt{3} = \sqrt{12\times 3} = \sqrt{36} = 6$

- $\dfrac{\sqrt{a}}{\sqrt{b}} = \sqrt{\dfrac{a}{b}}$ $(b>0)$

$\Rightarrow \dfrac{\sqrt{3}}{\sqrt{7}} = \sqrt{\dfrac{3}{7}}$ $\Rightarrow \dfrac{\sqrt{40}}{\sqrt{10}} = \sqrt{\dfrac{4\cancel{0}}{1\cancel{0}}} = \sqrt{4} = 2$

4) $\sqrt{k^2\times m} = k\sqrt{m}$

- $\sqrt{72} = \sqrt{36\times 2} = 6\sqrt{2}$
- $\sqrt{98} = \sqrt{49\times 2} = 7\sqrt{2}$
- $\sqrt{75} = \sqrt{25\times 3} = \sqrt{25}\cdot\sqrt{3} = 5\sqrt{3}$
- $\sqrt{63} = \sqrt{9\times 7} = \sqrt{9}\cdot\sqrt{7} = 3\sqrt{7}$

5) $a\sqrt{m} + b\sqrt{m} = (a+b)\sqrt{m}$

- $\sqrt{18} + \sqrt{8} = \sqrt{9\times 2} + \sqrt{4\times 2}$

$= 3\sqrt{2} + 2\sqrt{2} \Rightarrow \sqrt{2}(3+2)$

$\Rightarrow 5\sqrt{2}.$

6) $(\sqrt{a}+\sqrt{b})(\sqrt{a}-\sqrt{b}) = a-b$

As, $(A+B)(A-B) = A^2-B^2$

$(\sqrt{a}+\sqrt{b})(\sqrt{a}-\sqrt{b}) = (\sqrt{a})^2-(\sqrt{b})^2$

$= a-b.$

- $(\sqrt{8}+\sqrt{3})(\sqrt{8}-\sqrt{3}) \Rightarrow (\sqrt{8})^2-(\sqrt{3})^2$

$\Rightarrow (8^{1/2})^2 - (3^{1/2})^2$

$\Rightarrow 8 - 3 \Rightarrow 5$

$\sqrt{7} + \sqrt{3} \neq \sqrt{10}$

but

$\sqrt{7} \cdot \sqrt{3} = \sqrt{7 \times 3} = \sqrt{21}$

This rule is applicable only in case of multiplication & Division

7) $(\sqrt{a} + \sqrt{b})^2 = a + b + 2\sqrt{ab}$

$(\sqrt{a} + \sqrt{b})^2 = (\sqrt{a})^2 + (\sqrt{b})^2 + 2\sqrt{a} \cdot \sqrt{b}$

$= a + b + 2\sqrt{ab}$

8) $(\sqrt{a} - \sqrt{b})^2 = a + b - 2\sqrt{ab}$

$(\sqrt{a} - \sqrt{b})^2 = (\sqrt{a})^2 + (\sqrt{b})^2 - 2\sqrt{a} \cdot \sqrt{b}$

$= a + b - 2\sqrt{ab}$

Example,

$(\sqrt{2} + \sqrt{13})^2 \Rightarrow$

As,

$(a+b)^2 = a^2 + b^2 + 2ab$

$(\sqrt{2} + \sqrt{13})^2 = (\sqrt{2})^2 + (\sqrt{13})^2 + 2\sqrt{2} \cdot \sqrt{13}$

$= 2 + 13 + 2\sqrt{2 \times 13}$

$= 15 + 2\sqrt{26}$

$(\sqrt{2} - \sqrt{13})^2 \Rightarrow$

As,

$(a-b)^2 = a^2 + b^2 - 2ab$

$(\sqrt{2} - \sqrt{13})^2 = (\sqrt{2})^2 + (\sqrt{13})^2 - 2\sqrt{2} \cdot \sqrt{13}$

$= 2 + 13 - 2\sqrt{2 \times 13}$

$= 15 - 2\sqrt{26}$

$(2^6 \div 4^2) + (\sqrt{144} \div 6) = ?$

(A) 4 (B) 5 (C) 6 (D) 7 (E) 8

$\Rightarrow [2^6 \div (2^2)^2] + [(12^2)^{\frac{1}{2}} \div 6]$

$\Rightarrow (2^6 \div 2^4) + (12 \div 6)$

$\Rightarrow 2^{(6-4)} + 2 \Rightarrow 4 + 2 = 6$

$\{(3^4 \div 3^2) \times 3\} - \sqrt{81} = ?$

(A) 9 (B) 18 (C) 27 (D) 36 (E) 45

$\Rightarrow [(3^4 \div 3^2) \times 3^1] - 9$

$\Rightarrow 3^{(4-2+1)} - 9 \Rightarrow 3^3 - 9 \Rightarrow 27 - 9$

$= 18$

$(5^3 \div 25) + (\sqrt[3]{64} \times \sqrt{9}) = ?$

(A) 15 (B) 16 (C) 19 (D) 18 (E) 17

$\Rightarrow (5^3 \div 5^2) + ((64^{\frac{1}{3}}) \times (9)^{\frac{1}{2}})$

$\Rightarrow 5^{(3-2)} + (4 \times 3)$

$\Rightarrow 5 + 12 = 17$

$[(\sqrt{256} - 2^3) \times 3] \div 6 = ?$

(A) 4 (B) 5 (C) 2 (D) 3 (E) 6

$[(\sqrt{256} - 2^3) \times 3] \div 6$

$\Rightarrow \frac{(16-8) \times 3}{6} \Rightarrow \frac{8 \times 3}{6} = 4$

$\sqrt[4]{(81^3)} = ?$

(A) 9 (B) 27 (C) 81 (D) 243 (E) 729

$(81^3)^{\frac{1}{4}} \Longrightarrow (81^{\frac{1}{4}})^3$

$81 \rightarrow 9^2 \rightarrow (3^2)^2$

$\rightarrow 3^4$

$\Rightarrow [(3^4)^{\frac{1}{4}}]^3 = 3^3 = 27$

- $\sqrt[4]{16^3}$

$16 \rightarrow 4^2 \rightarrow (2^2)^2 \rightarrow 2^4$

$(16^3)^{\frac{1}{4}} \Rightarrow (16^{\frac{1}{4}})^3$

$\downarrow$

$[(2^{\cancel{4}})^{\cancel{\frac{1}{4}}}]^3$

$\Rightarrow (2)^3 = 8$

- $\sqrt[4]{625^3}$

$625 = 25^2 = (5^2)^2 = 5^4$

$(625^3)^{\frac{1}{4}} \Rightarrow (625^{\frac{1}{4}})^3$

$\downarrow$

$[(5^{\cancel{4}})^{\cancel{\frac{1}{4}}}]^3 = (5)^3$

$= 125$

$\sqrt{2^6 \times 3^3} = ?$

(A) 12√3 (B) 24√3 (C) 48√3 (D) 24√6 (E) 72√3

$\sqrt{2^6 \times 3^3} \Rightarrow \sqrt{2^6} \cdot \sqrt{3^3}$

$= (2^6)^{\frac{1}{2}} \cdot \sqrt{3 \times 3 \times 3}$

$= 2^3 \cdot \sqrt{3^2 \times 3} \Rightarrow 2^3 \cdot \sqrt{3^2} \cdot \sqrt{3}$

$\Rightarrow 8 \times 3 \times \sqrt{3}$

$= 24\sqrt{3}$

$125^? = \frac{1}{25}$

(A) −2/3 (B) −3/2 (C) 2/3 (D) 3/2 (E) −1/3

$125^x = \frac{1}{25} \Rightarrow (5^3)^x = \frac{1}{5^2}$

$5^{3x} = 5^{-2}$

$3x = -2$

$x = -\frac{2}{3}$

[Base same
sign "="
So, powers are equal]

$32^{\frac{3}{5}} - 8^{\frac{2}{3}} = ?$

(A) 2 (B) 12 (C) 6 (D) 8 (E) 4

$(32)^{\frac{3}{5}} - (8)^{\frac{2}{3}}$

$(2^5)^{\frac{3}{5}} - (2^3)^{\frac{2}{3}} \Rightarrow 2^3 - 2^2$

$= 8 - 4$

$= 4$

$2^1 = 2$	$2^6 = 64$
$2^2 = 4$	$2^7 = 128$
$2^3 = 8$	$2^8 = 256$
$2^4 = 16$	$2^9 = 512$
$2^5 = 32$	$2^{10} = 1024$

$64^{\frac{2}{3}} \div 16^{\frac{3}{4}} = ?$

(A) 1/2 (B) 1 (C) 2 (D) 4 (E) 8

$64^{\frac{2}{3}} \div 16^{\frac{3}{4}} \Rightarrow (4^3)^{\frac{2}{3}} \div (2^4)^{\frac{3}{4}}$

$\Rightarrow 4^2 \div 2^3$

$\Rightarrow 16 \div 8$

$\Rightarrow 2$

$4^1 = 4$
$4^2 = 16$
$4^3 = 64$
$4^4 = 256$

$\sqrt[3]{2^{10}} \div 2^{\frac{1}{3}} = ?$

(A) 4 (B) 6 (C) 8 (D) 10 (E) 16

$(2^{10})^{\frac{1}{3}} \div 2^{\frac{1}{3}} \Rightarrow 2^{\frac{10}{3}} \div 2^{\frac{1}{3}}$

$\Rightarrow 2^{\left(\frac{10}{3}-\frac{1}{3}\right)}$

$\Rightarrow 2^{\frac{9}{3}} \Rightarrow 2^3$

$= 8$

{ Base same
Sign "÷" Division
So, powers subtract. }

$\left(64^{\frac{1}{3}}\right)^{-2} \times 4^2 = ?$

(A) 1/4 (B) 1/2 (C) 1 (D) 2 (E) 4

$\left(64^{\frac{1}{3}}\right)^{-2} \times 4^2$

$\Rightarrow \left[(4^3)^{\frac{1}{3}}\right]^{-2} \times 4^2 \Rightarrow 4^{-2} \times 4^2$

$\Rightarrow \frac{1}{4^2} \times 4^2$

$\Rightarrow 1$

OR

$= 4^{-2} \times 4^2$

$= 4^{(-2+2)}$

$= 4^0$

$= 1$

$\left(9^{\frac{3}{2}} \div 27^{\frac{1}{3}}\right) + 2 = ?$

(A) 9 (B) 10 (C) 12 (D) 11 (E) 13

$= \left[(3^2)^{\frac{3}{2}} \div (3^3)^{\frac{1}{3}}\right] + 2$

$= (3^3 \div 3^1) + 2$

$= \left(\frac{27}{3}\right) + 2 \Rightarrow 9 + 2$

$= 11$

$3^1 = 3$
$3^2 = 9$
$3^3 = 27$
$3^4 = 81$

$\sqrt[3]{64^2} + \sqrt[4]{81^2} = ?$

(A) 20 (B) 21 (C) 24 (D) 25 (E) 27

$(64^2)^{\frac{1}{3}} + (81^2)^{\frac{1}{4}} \Rightarrow (64^{\frac{1}{3}})^2 + (81^{\frac{1}{4}})^2$

$\Rightarrow [(4^3)^{\frac{1}{3}}]^2 + [(3^4)^{\frac{1}{4}}]^2$

$\Rightarrow 4^2 + 3^2$

$\Rightarrow 16 + 9 \Rightarrow 25$

$(\sqrt[3]{8})^{-2} \times (\sqrt{16})^3 = ?$

(A) 4 (B) 8 (C) 16 (D) 32 (E) 64

$(8^{\frac{1}{3}})^{-2} \times ((16)^{\frac{1}{2}})^3 \Rightarrow [(2^3)^{\frac{1}{3}}]^{-2} \times [(4^2)^{\frac{1}{2}}]^3$

$\Rightarrow 2^{-2} \times 4^3$

$\Rightarrow 2^{-2} \times (2^2)^3$

$\Rightarrow 2^{-2} \times 2^6 \Rightarrow 2^{(6-2)} = 2^4 = 16$

$\left\{\left(16^{\frac{1}{2}} + 8^{\frac{1}{3}}\right)^2\right\} \div 25 = ?$

(A) 6/25 (B) 12/25 (C) 24/25 (D) 36/25 (E) 49/25

$\Rightarrow [(16^{\frac{1}{2}}) + (8^{\frac{1}{3}})]^2 \div 25$

$\left\{ \begin{array}{l} 4^2 = 16 \\ 2^3 = 8 \end{array} \right\}$

$\Rightarrow (4^{2\times\frac{1}{2}} + 2^{3\times\frac{1}{3}})^2 \div 25$

$\Rightarrow (4+2)^2 \div 25$

$\Rightarrow 6^2 \div 25 \Rightarrow \frac{36}{25}$ ✓

$16^{-2.25} = ?$

(A) 1/256 (B) 1/512 (C) 1/1024 (D) 512 (E) 1024

$16^{-9/4} \Rightarrow \frac{1}{16^{\frac{9}{4}}} \Rightarrow \frac{1}{(2^4)^{\frac{9}{4}}}$

$\Rightarrow \frac{1}{2^9} \Rightarrow \frac{1}{512}$

$2.25 = 2 + 0.25$

$= 2 + \frac{1}{4}$

$= \frac{8+1}{4} = \frac{9}{4}$

$(0.04)^{-\frac{1}{2}} = ?$

(A) 2 (B) 4 (C) 5 (D) 10 (E) 25

$\left(\frac{4}{100}\right)^{-\frac{1}{2}} \Rightarrow \left(\frac{1}{25}\right)^{-\frac{1}{2}}$

$\Rightarrow 25^{\frac{1}{2}}$

$= \sqrt{25} = 5$

{ Reciprocal and power sign change }

$81^{0.5} \times 81^{-0.25} = ?$

(A) 1 (B) 27 (C) 9 (D) 3 (E) 81

$\Rightarrow (81)^{0.5} \times (81)^{-0.25}$

$\Rightarrow (81)^{0.50-0.25}$

$\Rightarrow 81^{0.25} \Rightarrow 81^{\frac{1}{4}}$

$\Rightarrow 3^{4\times\frac{1}{4}} \Rightarrow 3^{1} \Rightarrow 3$

$\left(\frac{125}{27}\right)^{\frac{2}{3}} = ?$

(A) 5/3 (B) 25/9 (C) 9/25 (D) 10/9 (E) 5/9

$\left(\frac{125}{27}\right)^{\frac{2}{3}}$

$\left\{ \frac{125}{27} = \frac{5^3}{3^3} = \left(\frac{5}{3}\right)^3 \right\}$

$\Rightarrow \left[\left(\frac{5}{3}\right)^3\right]^{\frac{2}{3}} \Rightarrow \left(\frac{5}{3}\right)^{3\times\frac{2}{3}}$

$\Rightarrow \left(\frac{5}{3}\right)^2 \Rightarrow \frac{25}{9}$

$\sqrt{\frac{98}{72}} = ?$

(A) 7/3 (B) 7/6 (C) 14/3 (D) 6/7 (E) 3/7

$\sqrt{\frac{98}{72}} = \sqrt{\frac{49\times \cancel{2}}{36\times \cancel{2}}} = \frac{\sqrt{49}}{\sqrt{36}} = \frac{7}{6}$

$\left(\frac{16}{81}\right)^{-\frac{3}{4}} + \left(\frac{27}{125}\right)^{-\frac{2}{3}} = ?$

(A) 443/36 (B) 443/24 (C) 443/72 (D) 443/18 (E) None of these

$\Rightarrow \left(\frac{16}{81}\right)^{-\frac{3}{4}} + \left(\frac{27}{125}\right)^{-\frac{2}{3}} \Rightarrow \left(\frac{2^4}{3^4}\right)^{-\frac{3}{4}} + \left(\frac{3^3}{5^3}\right)^{-\frac{2}{3}}$

$\Rightarrow \left(\frac{2}{3}\right)^{4\times-\frac{3}{4}} + \left(\frac{3}{5}\right)^{3\times-\frac{2}{3}} \Rightarrow \left(\frac{2}{3}\right)^{-3} + \left(\frac{3}{5}\right)^{-2} \Rightarrow \left(\frac{3}{2}\right)^{3} + \left(\frac{5}{3}\right)^{2}$

$\Rightarrow \frac{27}{8} + \frac{25}{9} = \frac{243+200}{72} = \frac{443}{72}$

$\left(9^{-\frac{3}{2}} + 27^{-1}\right)^{-1} = ?$

(A) 9/2 (B) 27/2 (C) 9/4 (D) 27/4 (E) 81/2

$\left(\frac{1}{9} + \frac{1}{27}\right)^{-1} \Rightarrow \left(\frac{1}{27} + \frac{1}{27}\right)^{-1} = \left(\frac{2}{27}\right)^{-1}$

$= \frac{27}{2}$

$\left\{ 9^{\frac{3}{2}} = \left(9^{\frac{1}{2}}\right)^3 = \left(3^{2\times\frac{1}{2}}\right)^3 = 3^3 = 27 \right\}$

$\sqrt{3.24} = ?$

(A) 1.2 (B) 1.5 (C) 3.6 (D) 2.4 (E) 1.8

$\left(\frac{324}{100}\right)^{\frac{1}{2}} = \left(\frac{18^2}{10^2}\right)^{\frac{1}{2}}$

$= \left(\frac{18}{10}\right)^{2\times\frac{1}{2}} = \frac{18}{10}$ or 1.8

$\sqrt{0.81} + \sqrt[3]{0.027} = ?$

(A) 0.6 (B) 0.9 (C) 1.5 (D) 1.2 (E) 2.1

$\left(\frac{81}{100}\right)^{\frac{1}{2}} + \left(\frac{27}{1000}\right)^{\frac{1}{3}} \Rightarrow \left(\frac{9^2}{10^2}\right)^{\frac{1}{2}} + \left(\frac{3^3}{10^3}\right)^{\frac{1}{3}}$

$\Rightarrow \left(\frac{9}{10}\right)^{2\times\frac{1}{2}} + \left(\frac{3}{10}\right)^{3\times\frac{1}{3}} \Rightarrow \frac{9}{10} + \frac{3}{10} = \frac{12}{10}$ or 1.2

$(0.343)^{\frac{2}{3}} \times (0.49)^{-\frac{1}{2}} = ?$

(A) 0.7 (B) 0.5 (C) 0.9 (D) 1.4 (E) 0.07

$\Rightarrow \left(\frac{343}{1000}\right)^{\frac{2}{3}} \times \left(\frac{49}{100}\right)^{-\frac{1}{2}}$

$\Rightarrow \left(\frac{7^3}{10^3}\right)^{\frac{2}{3}} \times \left(\frac{7^2}{10^2}\right)^{-\frac{1}{2}} \Rightarrow \left(\frac{7}{10}\right)^{3\times\frac{2}{3}} \times \left(\frac{7}{10}\right)^{2\times-\frac{1}{2}}$

$\Rightarrow \left(\frac{7}{10}\right)^{2} \times \left(\frac{7}{10}\right)^{-1}$ [Reciprocal & sign change]

$\Rightarrow \frac{7}{10} \times \frac{\cancel{7}}{\cancel{10}} \times \frac{\cancel{10}}{\cancel{7}} = \frac{7}{10}$ or 0.7

$\left(a^{\frac{1}{2}} \times a^{\frac{3}{2}}\right) \div a^2 = ?$

(A) 1 (B) 1/a (C) a (D) √a (E) a^2

$\Rightarrow \left(a^{\frac{1}{2}} \times a^{\frac{3}{2}}\right) \div a^2$

$\Rightarrow \left(a^{\left(\frac{1}{2}+\frac{3}{2}\right)}\right) \div a^2$

$\Rightarrow \left(a^{\frac{4}{2}}\right) \div a^2$

$\Rightarrow a^2 \div a^2 = 1$

{ Base same
sign "X" Multiplication
So, sign of powers
is addition }

$\left(2^{\frac{5}{2}} \times 8^{\frac{1}{3}}\right) \div 4^{\frac{3}{2}} = ?$

(A) √2 (B) 2 (C) 1/√2 (D) 2√2 (E) √8

$\Rightarrow \left(2^{\frac{5}{2}} \times 8^{\frac{1}{3}}\right) \div 4^{\frac{3}{2}}$

$\Rightarrow \left(2^{\frac{5}{2}} \times (2^3)^{\frac{1}{3}}\right) \div (2^2)^{\frac{3}{2}}$

$\Rightarrow \left(2^{\frac{5}{2}} \times 2^1\right) \div 2^3$

$\Rightarrow 2^{\left(\frac{5}{2}+1\right)} \div 2^3 \Rightarrow 2^{\frac{7}{2}} \div 2^3$

$\Rightarrow 2^{\left(\frac{7}{2}-3\right)} \Rightarrow 2^{\left(\frac{7-6}{2}\right)} \Rightarrow 2^{\frac{1}{2}} \Rightarrow \sqrt{2}$

$\sqrt[3]{0.000512} = ?$

(A) 0.008 (B) 8 (C) 0.8 (D) 0.08 (E) 0.16

$$\sqrt[3]{\frac{512}{1000000}} = \left(\frac{8^3}{100^3}\right)^{\frac{1}{3}} = \left(\frac{8}{100}\right)^{3\times\frac{1}{3}}$$

$$= \frac{8}{100} \text{ or } 0.08$$

$0.64^{-\frac{3}{2}} \times 0.125^{-\frac{2}{3}} = ?$

(A) 125/16 (B) 125/8 (C) 125/4 (D) 125/32 (E) 25/16

$$\Rightarrow \left(\frac{64}{100}\right)^{-\frac{3}{2}} \times \left(\frac{125}{1000}\right)^{-\frac{2}{3}}$$

$$\Rightarrow \left[\left(\frac{8^2}{10^2}\right)^{\frac{1}{2}}\right]^{-3} \times \left[\left(\frac{5^3}{10^3}\right)^{\frac{1}{3}}\right]^{-2}$$

$$\Rightarrow \left(\frac{8}{10}\right)^{-3} \times \left(\frac{5}{10}\right)^{-2} \Rightarrow \left(\frac{10}{8}\right)^3 \times \left(\frac{10}{5}\right)^2 = \frac{1000}{512} \times \frac{\cancel{100}^{4}}{\cancel{25}_{1}} \Rightarrow \frac{125}{16}$$

$0.36^2 \times (0.216)^{\frac{1}{3}} = 0.6^?$

(A) 4 (B) 8 (C) 6 (D) 7 (E) 5

$$((0.6)^2)^2 \times (0.6^3)^{\frac{1}{3}} = 0.6^x$$

$$0.6^4 \times 0.6^1 = 0.6^x$$

$$(0.6)^{4+1} = 0.6^x$$

$$0.6^5 = 0.6^x$$

$$x = 5$$

$0.6^1 = 0.6$

$0.6^2 = 0.36$

$0.6^3 = 0.216$

$\sqrt[3]{0.216} \div \sqrt{0.49} = ?$

(A) 3/7 (B) 7/6 (C) 5/7 (D) 6/7 (E) 7/5

$$\Rightarrow \left(\frac{216}{1000}\right)^{\frac{1}{3}} \div \left(\frac{49}{100}\right)^{\frac{1}{2}} \Rightarrow \left(\frac{6^3}{10^3}\right)^{\frac{1}{3}} \div \left(\frac{7^2}{10^2}\right)^{\frac{1}{2}}$$

$$\Rightarrow \left(\frac{6}{10}\right)^{3\times\frac{1}{3}} \div \left(\frac{7}{10}\right)^{2\times\frac{1}{2}} \Rightarrow \frac{6}{10} \div \frac{7}{10} \Rightarrow \frac{6}{\cancel{10}} \times \frac{\cancel{10}}{7} = \frac{6}{7}$$

· Rationalisation means where we convert the denominator of a fraction from irrational to rational { Dividing and multiplying the root of denominator }

$(\sqrt{50} \times \sqrt{8}) \div \sqrt{2} = ?$

(A) 5√2 (B) 10√2 (C) 20√2 (D) 10√5 (E) 20

$\Rightarrow (\sqrt{50} \times \sqrt{8}) \div \sqrt{2}$

$\Rightarrow \sqrt{50 \times 8} \div \sqrt{2} \Rightarrow \sqrt{400} \div \sqrt{2}$

$\Rightarrow \sqrt{\frac{400}{2}} = \frac{20}{\sqrt{2}}$

{ Rationalization }

$\frac{20}{\sqrt{2}} \times \frac{\sqrt{2}}{\sqrt{2}} = \frac{20\sqrt{2}}{2}$

$= 10\sqrt{2}$ ✓

$(3\sqrt{18} + 5\sqrt{32}) \times \sqrt{2} = ?$

(A) 58 (B) 56 (C) 60 (D) 62 (E) 64

$(3\sqrt{18} + 5\sqrt{32}) \times \sqrt{2}$

$3\sqrt{18} \cdot \sqrt{2} + 5\sqrt{32} \cdot \sqrt{2}$

$3\sqrt{18 \times 2} + 5\sqrt{32 \times 2} \Rightarrow 3\sqrt{36} + 5\sqrt{64}$

$\Rightarrow 3 \times 6 + 5 \times 8$

$\Rightarrow 18 + 40 \Rightarrow 58$

$(\sqrt{75} \times \sqrt{12}) \div \sqrt{3} = ?$

(A) 10√3 (B) 5√3 (C) 15√3 (D) 30 (E) 10√6

$\Rightarrow (\sqrt{75} \times \sqrt{12}) \div \sqrt{3}$

$\Rightarrow \frac{\sqrt{25} \cdot \sqrt{3} \times \sqrt{4} \cdot \sqrt{3}}{\sqrt{3}} \Rightarrow \frac{5 \times \cancel{\sqrt{3}} \times 2\sqrt{3}}{\cancel{\sqrt{3}}}$

$\Rightarrow 10\sqrt{3}$

$\sqrt{180} + \sqrt{45} - \sqrt{20} = ?$

(A) 5√5 (B) 6√5 (C) 8√5 (D) 7√5 (E) 9√5

$\Rightarrow \sqrt{180} + \sqrt{45} - \sqrt{20}$

$\Rightarrow \sqrt{36 \times 5} + \sqrt{9 \times 5} - \sqrt{4 \times 5}$

$\Rightarrow 6\sqrt{5} + 3\sqrt{5} - 2\sqrt{5}$

$\Rightarrow \sqrt{5}(6 + 3 - 2) \Rightarrow \sqrt{5} \times 7 \Rightarrow 7\sqrt{5}$

$\sqrt[3]{250} \times \sqrt[3]{16} = ?$

(A) $10\sqrt[3]{4}$ (B) $10\sqrt[3]{2}$ (C) $5\sqrt[3]{32}$ (D) $20\sqrt[3]{2}$ (E) $4\sqrt[3]{10}$

$\Rightarrow \sqrt[3]{250} \times \sqrt[3]{16}$

$\Rightarrow \sqrt[3]{250 \times 16} \Rightarrow \sqrt[3]{4000}$

$\Rightarrow \sqrt[3]{1000 \times 4} \Rightarrow \sqrt[3]{1000} \times \sqrt[3]{4}$

$\Rightarrow 10\sqrt[3]{4}$

$(\sqrt{5} + 2\sqrt{20}) - \sqrt{80} = ?$

(A) √5 (B) 2√5 (C) 3√5 (D) 4√5 (E) 5√5

$\Rightarrow (\sqrt{5} + 2\sqrt{20}) - \sqrt{80}$

$\Rightarrow (\sqrt{5} + 2\sqrt{4 \times 5}) - \sqrt{16 \times 5}$

$\Rightarrow \sqrt{5} + (2 \times 2 \times \sqrt{5}) - (4 \times \sqrt{5})$

$\Rightarrow \sqrt{5}\,[1 + 4 - 4] \Rightarrow \sqrt{5}\,(1) \Rightarrow \sqrt{5}$

$\left(\frac{\sqrt{12}}{\sqrt{3}}\right) + \left(\frac{\sqrt{48}}{\sqrt{12}}\right) = ?$

(A) 4 (B) 2 (C) 5 (D) 3 (E) 6

$\Rightarrow \left(\frac{\sqrt{12}}{\sqrt{3}}\right) + \left(\frac{\sqrt{48}}{\sqrt{12}}\right)$

$\Rightarrow \frac{\sqrt{4 \times 3}}{\sqrt{3}} + \frac{\sqrt{12 \times 4}}{\sqrt{12}}$

$\Rightarrow \left(\frac{2\cancel{\sqrt{3}}}{\cancel{\sqrt{3}}}\right) + \left(\frac{2\cancel{\sqrt{12}}}{\cancel{\sqrt{12}}}\right) \Rightarrow 2 + 2$

$\Rightarrow 4$

$$\sqrt{58 + \sqrt{25 + \sqrt{81 + \sqrt{64} \times \sqrt{25}}}} = ?$$

(A) 6 (B) 7 (C) 8 (D) 9 (E) 10

$\Rightarrow \sqrt{58 + \sqrt{25 + \sqrt{81 + (8 \times 5)}}}$

$\Rightarrow \sqrt{58 + \sqrt{25 + \sqrt{81 + 40}}}$

$\Rightarrow \sqrt{58 + \sqrt{25 + 11}}$

$\Rightarrow \sqrt{58 + 6} = \sqrt{64} = 8$

$\sqrt{25} = 5$
$\sqrt{64} = 8$
$\sqrt{81} = 9$
$\sqrt{121} = 11$
$\sqrt{36} = 6$

$$\sqrt{74 + \sqrt{44 + \sqrt{13 + \sqrt{9} \times \sqrt{16}}}} = ?$$

(A) 6 (B) 7 (C) 8 (D) 9 (E) 10

$\Rightarrow \sqrt{74 + \sqrt{44 + \sqrt{13 + (3 \times 4)}}}$

$\Rightarrow \sqrt{74 + \sqrt{44 + \sqrt{25}}}$

$\Rightarrow \sqrt{74 + \sqrt{44 + 5}} \Rightarrow \sqrt{74 + \sqrt{49}}$

$\Rightarrow \sqrt{74 + 7} = \sqrt{81} = 9$

$\sqrt{16} = 4$
$\sqrt{9} = 3$
$\sqrt{25} = 5$
$\sqrt{49} = 7$
$\sqrt{81} = 9$

$$(\sqrt{6} + \sqrt{2})(\sqrt{6} - \sqrt{2}) + (\sqrt{3})^2 = ?$$

(A) 5 (B) 6 (C) 7 (D) 8 (E) 9

$(a+b)(a-b) = a^2 - b^2$

$(\sqrt{6}+\sqrt{2})(\sqrt{6}-\sqrt{2}) = (\sqrt{6})^2 - (\sqrt{2})^2 = 6 - 2 = 3$

As per Question,

$(\sqrt{6}+\sqrt{2})(\sqrt{6}-\sqrt{2}) + (\sqrt{3})^2$

$\Rightarrow 3 + (\sqrt{3})^2$

$\Rightarrow 3 + 3$

$\Rightarrow 6$

$(\sqrt{8} + \sqrt{18})^2 = ?$

(A) 50 (B) 25 (C) 75 (D) 100 (E) 125

$(a+b)^2 = a^2 + b^2 + 2ab$

$(\sqrt{8}+\sqrt{18})^2 = (\sqrt{8})^2 + (\sqrt{18})^2 + 2\sqrt{8}\cdot\sqrt{18}$

$= 8 + 18 + 2\sqrt{8\times 18}$

$= 8 + 18 + 2\sqrt{144}$ $\{\sqrt{144} = 12\}$

$= 8 + 18 + 2\times 12 \Rightarrow 26 + 24$

$\Rightarrow 50$

$(\sqrt{7} - \sqrt{5})^2 = ?$

(A) 12 + 2√35 (B) 12 – 2√35 (C) 2√35 – 12 (D) 2√35 + 12 (E) √35 – 12

$(a-b)^2 = a^2 + b^2 - 2ab$

$(\sqrt{7}-\sqrt{5})^2 = (\sqrt{7})^2 + (\sqrt{5})^2 - 2\sqrt{7}\cdot\sqrt{5}$

$= (7 + 5) - 2\sqrt{7\times 5}$

$= (12) - (2\sqrt{35})$

$= 12 - 2\sqrt{35}$

$(\sqrt{3} + \sqrt{5})^2 + (\sqrt{3} - \sqrt{5})^2 = ?$

(A) 8 (B) 12 (C) 16 (D) 18 (E) 20

$(a+b)^2 = a^2 + b^2 + 2ab$

$(a-b)^2 = a^2 + b^2 - 2ab$

$(\sqrt{3}+\sqrt{5})^2 + (\sqrt{3}-\sqrt{5})^2$

$[(\sqrt{3})^2 + (\sqrt{5})^2 + 2\sqrt{3}\cdot\sqrt{5}] + [(\sqrt{3})^2 + (\sqrt{5})^2 - 2\sqrt{3}\cdot\sqrt{5}]$

$(3 + 5 + 2\sqrt{15}) + (3 + 5 - 2\sqrt{15})$

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$$= (8 + 2\sqrt{15}) + (8 - 2\sqrt{15})$$
$$= 8 + 2\sqrt{15} + 8 - 2\sqrt{15}$$
$$= 16 + 2\sqrt{15} - 2\sqrt{15}$$
$$= 16$$

$\frac{3}{\sqrt{5}-\sqrt{2}} - \frac{3}{\sqrt{5}+\sqrt{2}} = ?$

(A) √2 (B) 2√2 (C) 3√2 (D) 4√2 (E) 6√2

$$\Rightarrow \frac{3}{\sqrt{5}-\sqrt{2}} - \frac{3}{\sqrt{5}+\sqrt{2}}$$ (Taking LCM)

$$\Rightarrow \frac{3(\sqrt{5}+\sqrt{2}) - 3(\sqrt{5}-\sqrt{2})}{(\sqrt{5}-\sqrt{2})(\sqrt{5}+\sqrt{2})}$$

$$\left\{ \begin{aligned} (a+b)(a-b) &= a^2 - b^2 \\ (\sqrt{5}+\sqrt{2})(\sqrt{5}-\sqrt{2}) &= (\sqrt{5})^2 - (\sqrt{2})^2 \\ &= 5-2 \\ &= 3 \end{aligned} \right\}$$

$$\Rightarrow \frac{\cancel{3\sqrt{5}} + 3\sqrt{2} - \cancel{3\sqrt{5}} + 3\sqrt{2}}{5-2} \Rightarrow \frac{3\sqrt{2}+3\sqrt{2}}{3} = \frac{6\sqrt{2}}{3} \Rightarrow 2\sqrt{2}$$

$\sqrt{19 + 6\sqrt{10}} - \sqrt{19 - 6\sqrt{10}} = ?$

(A) 2 (B) 4 (C) 8 (D) 6 (E) 10

$$(\sqrt{10}+3)^2 = (\sqrt{10})^2 + (3)^2 + 2\sqrt{10}\times 3$$
$$= 10 + 9 + 6\sqrt{10}$$
$$= 19 + 6\sqrt{10}.$$

$$\left\{ \begin{aligned} &\text{Using Identity} \\ &(a+b)^2 = a^2+b^2+2ab \\ &(a-b)^2 = a^2+b^2-2ab \end{aligned} \right\}$$

$$\Rightarrow \sqrt{19+6\sqrt{10}} - \sqrt{19-6\sqrt{10}}$$
$$\Rightarrow \sqrt{(\sqrt{10}+3)^2} - \sqrt{(\sqrt{10}-3)^2} \Rightarrow \sqrt{10}+3-(\sqrt{10}-3)$$
$$\Rightarrow \cancel{\sqrt{10}} + 3 - \cancel{\sqrt{10}} + 3$$
$$= 6$$

$(3\sqrt{5})(2\sqrt{15}) \div \sqrt{3} = ?$

(A) 30 (B) 20 (C) 10 (D) 40 (E) 50

$$\frac{3\sqrt{5} \times 2\sqrt{5\times 3}}{\sqrt{3}} \Rightarrow \frac{3\times\sqrt{5}\times 2\times\sqrt{5}\times\cancel{\sqrt{3}}}{\cancel{\sqrt{3}}}$$

$$\Rightarrow 3\times 2\times(\sqrt{5}\times\sqrt{5}) \Rightarrow 3\times 2\times 5 \Rightarrow 30$$

$(\sqrt{50} - \sqrt{8}) \div \sqrt{2} = ?$

(A) 1 (B) 3 (C) 2 (D) 4 (E) 5

$\Rightarrow (\sqrt{50} - \sqrt{8}) \div \sqrt{2}$

$\Rightarrow (\sqrt{25 \times 2} - \sqrt{4 \times 2}) \div \sqrt{2}$

$\Rightarrow \dfrac{5\sqrt{2} - 2\sqrt{2}}{\sqrt{2}} \Rightarrow \dfrac{\cancel{\sqrt{2}}(5-2)}{\cancel{\sqrt{2}}} \Rightarrow 3$

$\sqrt{2} \times \sqrt{3} \times \sqrt{12} \div \sqrt{2} = ?$

(A) 3 (B) 4 (C) 5 (D) 6 (E) 12

$\Rightarrow \dfrac{\sqrt{2} \times \sqrt{3} \times \sqrt{4 \times 3}}{\sqrt{2}}$

$\Rightarrow \dfrac{\cancel{\sqrt{2}} \times \sqrt{3} \times 2 \times \sqrt{3}}{\cancel{\sqrt{2}}} \Rightarrow (\sqrt{3} \times \sqrt{3}) \times 2$

$\Rightarrow 3 \times 2$

$\Rightarrow 6$

$(\sqrt{27} + \sqrt{12} - \sqrt{3}) \div \sqrt{3} = ?$

(A) 2 (B) 3 (C) 6 (D) 5 (E) 4

$\Rightarrow \dfrac{\sqrt{9 \times 3} + \sqrt{4 \times 3} - \sqrt{3}}{\sqrt{3}}$

$\Rightarrow \dfrac{3\sqrt{3} + 2\sqrt{3} - \sqrt{3}}{\sqrt{3}} \Rightarrow \dfrac{\cancel{\sqrt{3}}(3+2-1)}{\cancel{\sqrt{3}}}$

$\Rightarrow 4$

$(\sqrt[3]{54} \times \sqrt[3]{16}) \div \sqrt[3]{2} = ?$

(A) $3\sqrt[3]{2}$ ~~(B)~~ $6\sqrt[3]{2}$ (C) $12\sqrt[3]{2}$ (D) $6\sqrt[3]{4}$ (E) $2\sqrt[3]{54}$

$\Rightarrow (\sqrt[3]{54} \times \sqrt[3]{16}) \div \sqrt[3]{2}$

$\Rightarrow (\sqrt[3]{27 \times 2} \times \sqrt[3]{8 \times 2}) \div \sqrt[3]{2}$

$\Rightarrow \dfrac{3 \times \cancel{(2)^{1/3}} \times 2 \times (2)^{1/3}}{\cancel{(2)^{1/3}}} = 3 \times 2 \times (2)^{1/3}$

$= 6 \times 2^{\frac{1}{3}}$

$= 6\sqrt[3]{2}$

$\sqrt[3]{250} + \sqrt[3]{54} + \sqrt[3]{16} = ?$

(A) $5\sqrt[3]{2}$ (B) $8\sqrt[3]{2}$ ~~(C)~~ $10\sqrt[3]{2}$ (D) $12\sqrt[3]{2}$ (E) $15\sqrt[3]{2}$

$\Rightarrow (\sqrt[3]{125 \times 2}) + (\sqrt[3]{27 \times 2}) + (\sqrt[3]{8 \times 2})$

$\Rightarrow \sqrt[3]{5^3 \times 2} + \sqrt[3]{3^3 \times 2} + \sqrt[3]{2^3 \times 2}$

$\Rightarrow [5 \times (2)^{1/3}] + [3 \times (2)^{1/3}] + [2 \times (2)^{1/3}]$

$\Rightarrow 2^{\frac{1}{3}}[5 + 3 + 2] \Rightarrow 10\sqrt[3]{2}$

$\sqrt[3]{54} \times \sqrt{12} \div \sqrt{3} = ?$

(A) $3\sqrt[3]{2}$ ~~(B)~~ $6\sqrt[3]{2}$ (C) $9\sqrt[3]{2}$ (D) $12\sqrt[3]{2}$ (E) $18\sqrt[3]{2}$

$\Rightarrow \sqrt[3]{27 \times 2} \times \sqrt{4 \times 3} \div \sqrt{3}$

$\Rightarrow \dfrac{3 \times \sqrt[3]{2} \times 2 \times \cancel{\sqrt{3}}}{\cancel{\sqrt{3}}} \Rightarrow 3 \times 2 \times \sqrt[3]{2}$

$\Rightarrow 6 \times \sqrt[3]{2}$

$\Rightarrow 6\sqrt[3]{2}$

$(\sqrt{144} \times \sqrt{5}) \div (3 \times \sqrt{8} \times \sqrt{10}) = ?$

(A) 1 (B) 2 (C) 3 (D) 4 (E) 5

$\Rightarrow (\sqrt{144} \times \sqrt{5}) \div (3 \times \sqrt{8} \times \sqrt{10})$

$\Rightarrow (12 \times \sqrt{5}) \div (3 \times \sqrt{4 \times 2} \times \sqrt{2 \times 5}$

$\Rightarrow \dfrac{12 \times \cancel{\sqrt{5}}}{3 \times 2 \times \sqrt{2} \times \sqrt{2} \times \cancel{\sqrt{5}}} \Rightarrow \dfrac{12}{6 \times (\sqrt{2} \times \sqrt{2})} \Rightarrow \dfrac{12}{6 \times 2} \Rightarrow \dfrac{12}{12} = 1$

$(3\sqrt{18} + 2\sqrt{50}) \times \sqrt{2} = ?$

(A) 34 (B) 36 (C) 38 (D) 40 (E) 42

$\Rightarrow (3\sqrt{9 \times 2} + 2\sqrt{25 \times 2}) \times \sqrt{2}$

$\Rightarrow [(3 \times 3 \times \sqrt{2}) + (2 \times 5 \times \sqrt{2})] \times \sqrt{2}$

$\Rightarrow (9\sqrt{2} + 10\sqrt{2}) \times \sqrt{2}$

$\Rightarrow (9\sqrt{2} \cdot \sqrt{2}) + (10\sqrt{2} \cdot \sqrt{2}) \Rightarrow 9 \times 2 + 10 \times 2$

$\Rightarrow 18 + 20 \Rightarrow 38$

$(\sqrt[3]{54} + \sqrt[3]{16})^2 \div \sqrt[3]{4} = ?$

(A) 20 (B) 25 (C) 30 (D) 35 (E) 40

$\Rightarrow [\sqrt[3]{27 \times 2} + \sqrt[3]{8 \times 2}]^2 \div \sqrt[3]{4}$

$\Rightarrow [3 \times (2)^{\frac{1}{3}} + 2 \times (2)^{\frac{1}{3}}]^2 \div (2^2)^{\frac{1}{3}}$

$\Rightarrow \dfrac{[5 \times (2)^{\frac{1}{3}}]^2}{(2^2)^{\frac{1}{3}}} \Rightarrow \dfrac{(5)^2 \times (2)^{\frac{1}{3} \times 2}}{2^{2 \times \frac{1}{3}}} \Rightarrow \dfrac{25 \times \cancel{2^{2/3}}}{\cancel{2^{2/3}}}$

$\Rightarrow 25$